EFAMAT 200
(Efavirenz Tablets USP 200mg)

Label claim

EFAMAT Tablets USP 200 mg
Each film coated tablet contains
Efavirenz USP 200 mg

Product Description:
Peach colored oval shaped, film coated tablets, debossed with '3X' on one side and 'M' on other side.

List of Excipients:
Cellulose Microcrystalline, Croscarmellose sodium, Hydroxypropyl cellulose, Sodium lauryl Sulfate, Lactose monohydrate, Magnesium Stearate, Film coat {Hypromellose, Titanium dioxide, Polyethylene glycol 400, Iron oxide yellow, Iron oxide red}.

Therapeutic indications
Efavirenz is indicated in antiviral combination treatment of HIV 1 infected adults, adolescents and children.

Posology and method of administration

Adults: The recommended dosage of Efavirenz in combination with a protease inhibitor and/or nucleoside analogue reverse transcriptase inhibitors (NRTIs) is 600 mg orally, once daily. Efavirenz may be taken with or without food, as desired.

In order to improve the tolerability of nervous system side effects, bedtime dosing is recommended during the first two to four weeks of therapy and in patients who continue to experience these symptoms.(see SIDE EFFECTS).

Concomitant Antiretroviral Therapy: Efavirenz must be given in combination with other antiretroviral medications.

Adolescents and children (17 years and under): The recommended dose of Efavirenz incombination with a protease inhibitor and/or NRTIs for patients 17 years of age and under is described in Table 1. Efavirenz tablets should only be administered to children who are able to reliably swallow tablets. Efavirenz tablets may be taken with or without food, as desired. Efavirenz has not been adequately studied in children under the age of 3 years or children weighing less than 13 kg.

Table 1 Pediatric Dose to be Administered Once Daily

Body Weight (Kg)	**Efavirenz Tablets Dose (mg)**
13 to <15	200
15 to < 20	250
20 to < 25	300
25 to < 32.5	350

32.5 to < 40	400
≥40	600

Efamat Tablets are available in two strengths of 200mg and 600mg. They are not able to deliver all approved dose regimens of Efavirenz; other approved dosage forms and strengths of Efavirenz should be used in such cases.

Contraindications

Hypersensitivity to Efavirenz or to any of the excipients.

Efavirenz must not be used in patients with severe hepatic impairment (Child Pugh Grade C).

Efavirenz must not be administered concurrently with terfenadine, astemizole, cisapride, midazolam, triazolam, pimozide, bepridil, or ergot alkaloids (for example, ergotamine, dihydroergotamine, ergonovine, and methylergonovine) because competition for CYP3A4 by efavirenz could result in inhibition of metabolism and create the potential for serious and/or life-threatening undesirable effects [for example, cardiac arrhythmias, prolonged sedation or respiratory depression].
Herbal preparations containing St. John's wort (*Hypericum perforatum*) must not be used while taking efavirenz due to the risk of decreased plasma concentrations and reduced clinical effects of efavirenz.

Special warning and precautions for use

Special warning and precautions for use
Efavirenz must not be used as a single agent to treat HIV or added on as a sole agent to a failing regimen. The choice of new antiretroviral agent(s) to be used in combination with efavirenz should take into consideration the potential for viral cross resistance.

Coadministration of efavirenz with the fixed combination tablet containing efavirenz, emtricitabine, and tenofovir disoproxil fumarate, is not recommended.

Physicians should refer to the corresponding Summary of Product Characteristics.

QTc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Interaction with Other Medicaments). Consider alternatives to Efamat 200 when coadministered with a drug with a known risk of Torsade de Pointes or when administered to patients at higher risk of Torsade de Pointes.

Rash: Discontinued in patients developing severe rash associated with blistering, desquamation, mucosal involvement or fever.

Psychiatric symptoms: Patients with a prior history of psychiatric disorders appear to be at greater riskof these serious psychiatric adverse experiences. In particular severe depression, death by suicide, delusions and psychosis like behaviour.

Respective patients should contact their doctor immediately.

Nervous system symptoms:
Symptoms including dizziness, insomnia, somnolence, impaired concentration and abnormal dreaming are frequently reported in patients.

Late-onset neurotoxicity, including ataxia and encephalopathy (impaired consciousness, confusion, psychomotor slowing, psychosis, delirium), may occur months to years after beginning efavirenz therapy. Some events of late-onset neurotoxicity have occurred in patients with CYP2B6 genetic polymorphisms, which are associated with increased efavirenz levels despite standard dosing of Efamat 200. Patients presenting with signs and symptoms of serious neurologic adverse experiences should be evaluated promptly to assess the possibility that these events may be related to efavirenz use, and whether discontinuation of Efamat 200 is warranted.

Seizures: Convulsions have been observed rarely in patients receiving efavirenz, generallyin the presence of known medical history of seizures.

Caution must be taken in any patient with a history of seizures.

Effect of food: It is recommended that Efavirenz be taken on an empty stomach, preferably at bedtime.

Immune Reactivation Syndrome : In HIV infected patients with severe immune deficiency at the time of institution of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of CART. Relevant examples are cytomegalovirus retinitis, generalised and/or focal mycobacterial infections, and pneumonia caused by Pneumocystis jiroveci (formerly known as Pneumocystis carinii). Any inflammatory symptoms should be evaluated and treatment instituted when necessary.

Lipodystrophy and metabolic abnormalities: Combination antiretroviral therapy has been associated with the redistribution of body fat (lipodystrophy) in HIV patients. The long term consequences of these events are currently unknown.

Osteonecrosis: Patients should be advised to seek medical advice if they experience joint aches and pain, joint stiffness or difficulty in movement.

Lactose: This medicinal product contains 256 mg which not likely to induce symptoms of lactose intolerance.

Efavirenz film coated tablets are unsuitable for individuals with the rare hereditary disorders of galactosaemia. Individuals with these conditions may take efavirenz oral solution, which is free from lactose.

Special populations:
Liver disease: Caution must be exercised in administering efavirenz to patients with mild to moderate liver disease. Patients should be monitored carefully for dose related adverse events, especially nervous system symptoms. Laboratory tests should be performed to evaluate their liver disease at periodic intervals. Efavirenz is contraindicated in patients with severe hepatic impairment. Patients with chronic hepatitis B or C and treated with combination antiretroviral therapy are at increased risk for severe and potentially fatal hepatic adverse events. Patients with pre-existing liver dysfunction including chronic active hepatitis have an increased frequency of liver function abnormalities during combination antiretroviral therapy and should be monitored according to standard practice. If there is evidence of worsening liver disease or persistent elevations of serum transaminases to greater than 5 times the upper limit of the normal range. In such patients, interruption or discontinuation of treatment must be considered.
In patients treated with other medicinal products associated with liver toxicity, monitoring of liver enzymes is also recommended. In case of concomitant antiviral therapy for hepatitis B or C, please refer also to the relevant product information for these medicinal products.

Renal insufficiency: The impact of renal impairment on efavirenz elimination should be minimal. There is no experience in patients with severe renal failure and close safety monitoring is recommended in this population.

Elderly: Insufficient data to determine whether they respond differently than younger patients.

Children: Efavirenz has not been evaluated in children below 3 years of age or who weigh less than 13 kg.

Effects on ability to drive and use machines
Efavirenz has not been specifically evaluated for possible effects on the ability to drive a car or operate machinery.
Efavirenz may cause dizziness, impaired concentration, and/or somnolence. Patients should be instructed that if they experience these symptoms they should avoid potentially hazardous tasks such as driving or operating machinery.

Interaction with other medicinal products and other forms of interaction
Efavirenz is an inducer of CYP3A4 and an inhibitor of some CYP isozymes including CYP3A4. Other compounds that are substrates of CYP3A4 may have decreased plasma concentrations when co-administered with efavirenz. Efavirenz exposure may also be altered when given with medicinal products or food (for example, grapefruit juice) which affect CYP3A4 activity.

QT Prolonging Drugs
There is limited information available on the potential for a pharmacodynamic interaction between Efamat 200 and drugs that prolong the QTc interval. QTc prolongation has been observed with the use of efavirenz (see Section Pharmacodynamics and Section Warnings and Precautions). Consider alternatives to Efamat 200 when coadministered with a drug with a known risk of Torsade de Pointes.

Contraindications of concomitant use
Efavirenz must not be administered concurrently with terfenadine, astemizole, cisapride, midazolam, triazolam, pimozide, bepridil, or ergot alkaloids (for example, ergotamine, dihydroergotamine, ergonovine, and methylergonovine) since inhibition of their metabolism may lead to serious, life-threatening events.

St. John's wort (*Hypericum perforatum*): Co-administration of efavirenz and St. John's wort or herbal preparations containing St. John's wort is contraindicated. Plasma levels of efavirenz can be reduced by concomitant use of St. John's wort due to induction of drug metabolising enzymes and/or transport proteins by St. John's wort. If a patient is already taking St. John's wort, stop St. John's wort, check viral levels and if possible efavirenz levels. Efavirenz levels may increase on stopping St. John's wort and the dose of efavirenz may need adjusting. The inducing effect of St. John's wort may persist for at least 2 weeks after cessation of treatment.

Other interactions
Interactions between efavirenz and protease inhibitors, antiretroviral agents other than protease inhibitors and other non-antiretroviral medicinal products are listed in Table 2 below.

Table 2: Interactions between efavirenz and other medicinal products

Medicinal product by therapeutic areas (dose)	Recommendation concerning coadministration with efavirenz
ANTI-INFECTIVES	
Antiretrovirals	
Protease inhibitors (PI)	
Atazanavir/ ritonavir/Efavirenz (400 mg once daily/100mg once daily/600mg once daily, all administered with food) Atazanavir/ritonavir/Efavirenz (400 mg once daily/200mg daily/600 mg once daily, all administered with food)	Co-administration of efavirenz with atazanavir/ritonavir is not recommended. If co-administration of atazanavir with an NNRTI is required, an increase in the dose of both atazanavir and ritonavir to 400mg and 200mg, respectively, in combination efavirenz could be considered with close clinical monitoring.
Fosamprenavir/ritonavir/Efavirenz (700 mg twice daily/100 mg twice daily/600 mg once daily)	No dosage adjustment is necessary for any of these medicinal products. See also ritonavir row below.
Fosamprenavir/Nelfinavir/Efavirenz Fosamprenavir/Saquinavir/Efavirenz	No dosage adjustment is necessary for any of these medicinal products. Not recommended as the exposure to both PIs is expected to be significantly decreased.
Indinavir/Efavirenz (800 mg q8h/200 mg once daily)	While the clinical significance of decreased indinavir concentrations has not been established, the magnitude of the observed pharmacokinetics interaction should be taken into consideration when choosing a regimen containing both efavienz and indinavir.
Indinavir/ritonavir/Efavirenz (800 mg twice daily/100 mg twice daily/600 mg once daily)	No dosage adjustment is necessary for efavirenz when given with indinavir or indinavir/ritonavir. See also ritonavir row below.
Lopinavir/ritonavir soft capsules or oral solution/Efavirenz Lopinavir/ritonavir tablets/Efavirenz (400/100 mg twice daily/600 mg once daily) (500/125 mg twice daily/600 mg once daily)	With efavirenz, an increase of the lopinavir/ritonavir soft capsule or oral solution doses by 33% should be considered (4 capsules/~6.5 ml twice daily instead of 3 capsules/5 ml twice daily). Caution is warranted since this dosage adjustment might be insufficient in some patients. The dosage of lopinavir/ritonavir tablets should be increased to 500/125 mg twice daily when co- administered with efavirenz 600 mg once daily. See also ritonavir row below.
Nelfinavir/Efavirenz (750 mg q8h/600 mg once daily)	No dosage adjustment is necessary for either medicinal product.
Ritonavir/Efavirenz (500 mg twice daily/600 mg once daily)	When using efavirenz with low dose ritonavir, the possibility of an increase in the incidence of efavirenz-associated adverse events should be considered, due to possible pharmacodymic interaction.
Saquinavir/ritonavir/Efavirenz	No data are available to make a dose recommendation. See also ritonavir row above. Use of efavirenz in combination with saquinavir as the sole protease inhibitor is not recommended.
NRTIs and NNRTIs	

NRTIs/Efavirenz	No dosage adjustment is necessary for either medicinal product.
NNRTIs/Efavirenz	Since use of two NNRTIs proved not beneficial in terms of efficacy and safety, co-administration of efavirenz and another NNRTI is not recommended.
Antibiotics	
Azithromycin/Efavirenz (600 mg single dose/400 mg once daily)	No dosage adjustment is necessary for either medicinal product.
Clarithromycin/Efavirenz (500 mg q12h/400 mg once daily)	The clinical significance of these changes in clarithromycin plasma levels is not known. Alternatives to clarithromycin (e.g. azithromycin) may be considered. No dosage adjustment is necessary for efavirenz.
Other macrolide antibiotics (e.g.,erythromycin)/Efavirenz	No data are available to make a dose recommendation.
Antimycobacterials	
Rifabutin/Efavirenz (300 mg once daily/600 mg once daily)	The daily dose of rifabutin should be increased by 50% when administered with efavirenz. Consider doubling the rifabutin dose in regimens where rifabutin is given 2 or 3 times a week in combination with efavirenz.
Rifampicin/Efavirenz (600 mg once daily/600 mg once daily)	When taken with rifampicin, increasing efavirenz daily dose to 800 mg may provide exposure similar to a daily dose of 600 mg when taken without rifampicin. The clinical effect of this dose adjustment has not been adequately evaluated. Individual tolerability and virological response should be considered when making the dose adjustment. No dosage adjustment is necessary for rifampicin.
Antifungals	
Itraconazole/Efavirenz (200 mg q12h/600 mg once daily)	Since no dose recommendation for Itraconazole can be made, alternative antifungal treatment should be considered.
Voriconazole/Efavirenz (200 mg twice daily/400 mg once daily) Voriconazole/Efavirenz (400 mg twice daily/300 mg once daily)	When efavirenz is co-administered with voriconazole, the voriconazole maintenance dose must be increased to 400 mg twice daily and the efavirenz dose must be reduced by 50%, i.e., to 300 mg once daily. When treatment with voriconazole is stopped, the initial dosage of efavirenz should be restored.

Fluconazole/Efavirenz (200 mg once daily/400 mg once daily)	No dosage adjustment is necessary for either medicinal product.
Ketoconazole and other imidazole antifungals	No data are available to make a dose recommendation.
ACID REDUCING AGENTS	
Aluminium hydroxide-magnesium hydroxide-simethicone antacid/Efavirenz(30 mL single dose/400 mg single dose) Famotidine/Efavirenz (40 mg single dose/400 mg single dose)	Co-administration of efavirenz with medicinal products that after gastric pH would not be expected to affect efavirenz absorption.
ANTIANXIETY AGENTS	
Lorazepam/Efavirenz (2 mg single dose/600 mg once daily)	No dosage adjustment is necessary for either medicinal product.
ANTICONVULSANTS	
Carbamazepine/Efavirenz (400 mg once daily/600 mg once daily)	No dosage recommendation can be made. Anticonvulsant should be considered. Carbamazepine plasma levels periodically.
Phenytoin, Phenobarbital, and other anticonvulsants that are substrates of CYP450 isoenzymes	When efavirenz is coadministered with an anticonvulsant that is a substrate of CYP450 isoenzymes, periodic monitoring of anticonvulsant levels should be conducted.
Vigabatrin/Efavirenz Gabapentin/Efavirenz	No dosage adjustment is necessary for any of these medicinal products.
ANTIDEPRESSANTS	

Selective Serotonin Reuptake Inhibitors (SSRIs)	
Sertraline/Efavirenz (50 mg once daily/600 mg once daily)	Sertraline dose increases should be guided by clinical response. No dosage adjustment is necessary for efavirenz.
Paroxetine/Efavirenz (20 mg once daily/600 mg once daily)	No dosage adjustment is necessary for either medicinal product.
Fluoxetine/Efavirenz	No dosage adjustment is necessary for either medicinal product.
ANTIHISTAMINES	
Cetirizine/Efavirenz (10 mg single dose/600 mg once daily)	No dosage adjustment is necessary for either medicinal product.
CARDIOVASCULAR AGENTS	
Calcium Channel Blockers	
Diltiazem/Efavirenz (240 mg once daily/600 mg once daily)	Dose adjustments of diltiazem should be guided by clinical response (refer to the Summary of Product diltiazem). No dosage adjustment is necessary for efavirenz.
Verapamil, Felodipine, Nifedipine and Nicardipine	Dose adjustments of calcium channel blockers should be guided by clinical response (refer to the Summary Characteristics for the calcium channel blocker).
LIPID LOWERING MEDICINAL PRODUCTS	
HMG Co-A Reductase Inhibitors	
Atorvastatin/Efavirenz (10 mg once daily/600 mg once daily)	Cholesterol levels should be periodically monitored. Adjustment of atorvastatin may be required (refer to the summary of Product atorvastatin). No dosage adjustment is necessary for efavirenz.
Pravastatin/Efavirenz (40 mg once daily/600 mg once daily)	Cholesterol levels should be periodically monitored. Adjustment of pravastatin may be required (refer to the Summary of product pravastatin). No dosage adjustment is necessary for efavirenz.
Simvastatin/Efavirenz (40 mg once daily/600 mg once daily)	Cholesterol levels should be periodically monitored. Dosage adjustment of simvastatin may be required (refer to the Summary of product characteristics for simvastatin). No dosage adjustment is necessary for efavirenz.
HORMONAL CONTRACEPTIVES	
Ethinyloestradiol/Efavirenz (50µ single dose/400mg once daily)	The potential interaction of efavirenz with oral contraceptives has not been fully characterised. A reliable method of barrier contraception must be used in addition to oral contraceptives.
IMMUNOSUPPRESSANTS	
Immunosuppressants metabolized by CYP3A4 (eg, cyclosporine, tacrolimus, sirolimus)/Efavirenz	Dose adjustments of the immunosuppressant may be required. Close monitoring of immunosuppressant concentrations for at least 2 weeks (until stable concentrations recommended when starting or stopping treatment with efavirenz.

OPIOIDS	
Methadone/Efavirenz (stable maintenance, 35-100mg once daily/600 mg once daily)	Patients should be monitored for signs methadone dose increased as required to alleviate withdrawal symptoms.

Pregnancy and lactation
Efavirenz should not be used during pregnancy unless there are no other appropriate treatment options.

Women of childbearing potential: Pregnancy should be avoided in women treated with efavirenz. Barrier contraception should always be used in combination with other methods of contraception (for example, oral or other hormonal contraceptives). Because of the long half-life of efavirenz, use of adequate contraceptive measures for 12 weeks after discontinuation of efavirenz is recommended. Women of childbearing potential should undergo pregnancy testing before initiation of efavirenz.

Pregnancy: There are no adequate and well-controlled studies of efavirenz in pregnant women. In postmarketing experience through an antiretroviral pregnancy registry, more than 200 pregnancies with first-trimester exposure to efavirenz as part of a combination antiretroviral regimen have been reported with no specific malformation pattern. Retrospectively in this registry, a small number of cases of neural tube defects, including meningomyelocele, have been reported but causality has not been established. Studies in animals have shown reproductive toxicity including marked teratogenic effects.

Lactation: Studies in rats have demonstrated that efavirenz is excreted in milk reaching concentrations much higher than those in maternal plasma. It is not known whether efavirenz is excreted in human milk. Since animal data suggest that the substance may be passed into breast milk, it is recommended that mothers taking efavirenz do not breast feed their infants. It is recommended that HIV infected women do not breast feed their infants under any circumstances in order to avoid transmission of HIV.

Undesirable effects
The most frequently reported treatment-related undesirable effects of at least moderate severity reported in patients were rash, dizziness, nausea, headache and fatigue. The most notable undesirable effects associated with efavirenz are rash and nervous system symptoms. The administration of Efavirenz with food may increase efavirenz exposure and may lead to an increase in the frequency of undesirable effects.

Long-term use of efavirenz was not associated with any new safety concerns.
Rash: Rashes are usually mild to moderate maculopapular skin eruptions that occur within the first two weeks of initiating therapy with efavirenz. In most patients rash resolves with continuing therapy with efavirenz within one month. Efavirenz can be reinitiated in patients interrupting therapy because of rash. Use of appropriate antihistamines and/or corticosteroids is recommended when efavirenz is restarted.

Experience with efavirenz in patients who discontinued other antiretroviral agents of the NNRTI class is limited

Psychiatric symptoms: Serious psychiatric adverse experiences have been reported in patients treated with efavirenz. The specific serious psychiatric events are detailed hereafter:
- severe depression
- suicidal ideation
- non-fatal suicide attempts
- aggressive behaviour
- paranoid reactions
- manic reactions

Patients with a history of psychiatric disorders appear to be at greater risk of these serious psychiatric adverse experiences with the frequency of each of the above events ranging from 0.3% for manic reactions to 2.0% for both severe depression and suicidal ideation. There have also been post marketing reports of death by suicide, delusions and psychosis-like behaviour.

Nervous system symptoms: The frequently reported undesirable effects in patients receiving 600 mg efavirenz with other antiretroviral agents included, but were not limited to: dizziness, insomnia, somnolence, impaired concentration and abnormal dreaming. Nervous system symptoms of moderate-to- severe intensity were experienced by patients receiving control regimens.

Nervous system symptoms usually begin during the first one or two days of therapy and generally resolve after the first 2 -4 weeks.

Nervous system symptoms may occur more frequently when efavirenz is taken concomitantly with meals possibly due to increased efavirenz plasma levels. Dosing at bedtime seems to improve the tolerability of these symptoms and can be recommended during the first weeks of therapy and in patients who continue to experience these symptoms. Dose reduction or splitting the daily dose has not been shown to provide benefit.

Immune system disorders
Uncommon: hypersensitivity

Psychiatric disorders
Common: anxiety, depression
Uncommon: affect lability, aggression, euphoric mood, hallucination, mania, paranoia, suicide attempt, suicide ideation

Nervous system disorders

Common: abnormal dreams, disturbance in attention, dizziness, headache, insomnia, somnolence
Uncommon: agitation, amnesia, ataxia, coordination abnormal, confusional state, convulsions, thinking abnormal

Eye disorders

Uncommon: vision blurred

Ear and labyrinth disorders
Uncommon: vertigo

Gastrointestinal disorders
Common: abdominal pain, diarrhoea, nausea, vomiting
Uncommon: pancreatitis acute

Hepatobiliary disorders

Uncommon: hepatitis acute

Skin and subcutaneous tissue disorders
Very common: rash
Common: pruritus
Uncommon: erythema multiforme, Stevens-Johnson syndrome

General disorders and administration site conditions
Common: fatigue

Reproductive system and breast disorders

Uncommon: gynaecomastia

Immune Reactivation Syndrome: In HIV infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise.

Lipodystrophy and metabolic abnormalities: Combination antiretroviral therapy has been associated with redistribution of body fat (lipodystrophy) in HIV patients including the loss of peripheral and facial subcutaneous fat, increased intra-abdominal and visceral fat, breast hypertrophy and dorsocervical fat accumulation (buffalo hump).

Combination antiretroviral therapy has been associated with metabolic abnormalities such as hypertriglyceridaemia, hypercholesterolaemia, insulin resistance, hyperglycaemia and hyperlactataemia.

Osteonecrosis: Cases of osteonecrosis have been reported, particularly in patients with generally acknowledged risk factors, advanced HIV disease or long-term exposure to combination antiretroviral therapy (CART). The frequency of this is unknown.

Cannabinoid test interaction: Efavirenz does not bind to cannabinoid receptors. False positive urine cannabinoid test results have been reported in uninfected volunteers who received efavirenz. False positive test results have only been observed with the CEDIA DAU Multi-Level THC assay, which is used for screening, and have not been observed with other cannabinoid assays tested including tests used for confirmation of positive results.

Post marketing experience

Encephalopathy

Post marketing experience with efavirenz has shown the following additional adverse events to occur in association with efavirenz-containing antiretroviral treatment regimens: delusion, gynaecomastia, hepatic failure, neurosis, photoallergic dermatitis, psychosis and completed suicide.

Adolescents and children: Undesirable effects in children were generally similar to those of adult patients.

Rash was reported more frequently in children and was more often of higher grade than in adults.

Prophylaxis with appropriate antihistamines prior to initiating therapy with efavirenz in children may be considered. Although nervous system symptoms are difficult for young children to report, they appear to be less frequent in children and were generally mild.

Overdose

Some patients accidentally taking 600 mg twice daily have reported increased nervous system symptoms. One patient experienced involuntary muscle contractions.

Treatment of overdose with efavirenz should consist of general supportive measures, including monitoring of vital signs and observation of the patient's clinical status. Administration of activated charcoal may be used to aid removal of unabsorbed efavirenz. There is no specific antidote for overdose with efavirenz. Since efavirenz is highly protein bound, dialysis is unlikely to remove significant quantities of it from blood.

Pharmacodynamic properties

Pharmacotherapeutic group:

Efavirenz is a selective non-nucleoside reverse transcriptase inhibitor of human immunodeficiency virus type 1 (HIV-1). Efavirenz is a non-competitive inhibitor of HIV-1 reverse transcriptase (RT) with respect to template, primer or nucleoside triphosphates, with a small component of competitive inhibition. HIV-2RT and human cellular DNA polymerases α, β, γ and δ are not inhibited by concentrations of efavirenz well in excess of those achieved clinically.

Cardiac Electrophysiology

The effect of Efamat 200 on the QTc interval was evaluated in an open-label, positive and placebo controlled, fixed single sequence 3-period, 3-treatment crossover QT study in 58 healthy subjects enriched for CYP2B6 polymorphisms. The mean Cmax of efavirenz in subjects with CYP2B6 *6/*6 genotype following the administration of 600mg daily dose for 14 days was 2.25-fold the mean Cmax observed in subjects with CYP2B6 *1/*1 genotype. A positive relationship between efavirenz concentration and QTc prolongation was observed. Based on the concentration-QTc relationship, the mean QTc prolongation andits upper bound 90% confidence interval are 8.7 ms and 11.3 ms in subjects with CYP2B6*6/*6 genotype following the administration of 600mg daily dose for 14 days. (see Section Warnings and Precautions & Section Interaction with Other Medicaments).

Pharmacokinetic properties

Absorption

Peak efavirenz plasma concentrations were attained by 5 hours following single oral doses of 100 mg to 1600 mg administered to uninfected volunteers. Dose-related increases in Cmax and AUC were seen for doses up to 1600 mg; the increases were less than proportional suggesting diminished absorption at higher doses. Time to peak plasma concentrations (3-5 hours) did not change following multiple dosing and steady-state plasma concentrations were reached in 6-7 days.

Effect of Food on Oral Absorption

Efavirenz can be administered with or without food.

Distribution

Efavirenz is highly bound (approximately 99.5-99.75%) to human plasma proteins, predominantly albumin.

Metabolism

Efavirenz is principally metabolized by the cytochrome P450 system to hydroxylated metabolites with subsequent glucuronidation of these hydroxylated metabolites. These metabolites are essentially inactive against HIV-1. It is reported that CYP3A4 and CYP2B6 are the major isozymes responsible for efavirenz metabolism.

Efavirenz has been shown to induce P450 enzymes, resulting in the induction of its own metabolism.

Elimination

Efavirenz has a relatively long terminal half-life of 52 to 76 hours after single doses and 40-55 hours after multiple doses. Approximately 14-34% of a radiolabeled dose of efavirenz was recovered in the urine and less than 1% of the dose was excreted in urine as unchanged efavirenz.

Shelf life 36 Months

Special precautions for storage

Do not store above 30°C. Store in the original container.

Pack
HDPE bottle of 30's.

Product Registration Holder:
Pahang Pharmacy Sdn. Bhd.
Lot 5979, Jalan Teratai, 5 ½ Miles off Jalan Meru, 41050 Klang, Selangor, MALAYSIA.

Manufactured By:
Mylan Laboratories Limited
F-4 & F-12, MIDC, Malegaon, Sinnar, Nashik - 422 113, Maharashtra, INDIA

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